

A Survey and New Perspective of Sensing in the Dark for Intelligent Transportation Systems

Yiting Wang^{ID}, *Graduate Student Member, IEEE*, Boda Li^{ID}, Zixiang Wei^{ID}, *Graduate Student Member, IEEE*, Anima Rahman, Daniel Gummadi^{ID}, Haonan Zhao^{ID}, and Valentina Donzella^{ID}, *Senior Member, IEEE*

Abstract—Dark or low-luminance environments pose significant challenges for intelligent transportation systems (ITS), potentially affecting perception, decision-making, and safety. In this context, the concept of Vehicular Sensing in the Dark (DarkSensing) is introduced as the use of sensing technologies and algorithms to enable vehicles to perceive, communicate and navigate under dark conditions. It encompasses both individual sensor capabilities and cooperative sensing strategies across all vehicle automation levels. This survey provides the first comprehensive and structured overview of DarkSensing in ITS, focusing on five perception sensor modalities: cameras, LiDAR, RADAR, thermal sensors, and event-based sensors. Specifically, the survey systematically examines the challenges and recent advancements in sensor hardware, perception algorithms, data enhancement methods, and collaborative sensing frameworks. Furthermore, we highlight critical open issues with case studies and outline future research directions. This work establishes the conceptual foundation for DarkSensing and aims to guide researchers and practitioners toward robust ITS performance in dark conditions. Project link: <https://github.com/Warwick-Jocelyn/DarkSensing>

Index Terms—Dark perception, perception sensors, intelligent transportation systems.

I. INTRODUCTION

OVER the past decade, automated driving (AD) technologies have seen significant advancements due to the collaborative efforts of governments, researchers, and industries. Advanced driver assistance systems (ADASs) and AD systems have been deployed in vehicles, driven by the rapid evolution of hardware and software. While the ambitious vision of fully autonomous vehicles aligns with SAE Levels 4 and 5, most commercially deployed automated

systems currently operate at Level 2, as higher levels face limited real-world deployment due to regulatory, safety, and operational constraints,¹ as well as the challenges posed by validating and verifying countless driving scenarios under all conditions.

Among the various adverse environmental conditions, dark environments, ranging from low-light to night-light and extreme-dark scenarios, pose a significant threat to vehicular sensing systems. Such conditions are unavoidable in real-world operation and introduce challenges, particularly for imaging sensors, including reduced contrast, amplified noise, and colour distortions. These factors can severely degrade vision-based perception quality and compromise downstream tasks such as detection, segmentation, and tracking. The need for robust dark perception is increasingly urgent. An analysis of traffic crash reports involving autonomous vehicles (AVs) in California found that driving in poor lighting conditions increased the risk of road traffic injuries by more than **six times** [1]. Moreover, although only about 25% of the total driving occurs at night, **over 50%** of fatal traffic accidents happen during nighttime conditions [2], [3], [4]. Furthermore, AV perception systems suffer substantial performance drops in dark environments, for example, commonly used object detectors exhibit accuracy declines **exceeding 40%** on dark-scene benchmarks [5], [6]. Decreased visibility, prolonged glare recovery times, and difficulties in spotting vulnerable road users, especially cyclists and pedestrians who wear dark clothes, contribute to the heightened risk during night driving [7]. These findings underscore the need for specialised sensing strategies and comprehensive evaluation frameworks in various dark scenarios, motivating our proposal of the DarkSensing paradigm.

Many stakeholders aim to reduce the cost of ADAS systems by implementing pure vision-based solutions. However, current automotive Complementary Metal-Oxide-Semiconductor (CMOS) camera sensors do not guarantee their full functionality in dark environments [10]. An alternative is the use of night vision systems, such as thermal infrared cameras, which detect heat signatures emitted by objects such as pedestrians and animals, enabling detection even in complete darkness. Another approach can be to incorporate redundant sensors and use sensor fusion to improve system robustness, which, however, adds complexity, cost, and weight to vehicles. With

Received 22 December 2024; revised 19 May 2025; accepted 9 July 2025. Date of publication 6 October 2025; date of current version 3 November 2025. Co-funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Climate, Infrastructure and Environment Executive Agency (CINEA). Neither the European Union nor the Granting authority can be held responsible for them. Project Grant no. 101069576. UK participants in this project are co-funded by Innovate UK under contract no. 10045139. Swiss participants in this project are co-funded by the Swiss State Secretariat for Education, Research and Innovation (SERI) under contract no. 22.00123. The work was partially supported by High-Value Manufacturing CATAPULT, the Centre for Doctoral Training to Advance the Deployment of Future Mobility Technologies (CDT FMT), and the China Scholarship Council. The Associate Editor for this article was S.-H. Kong. (*Corresponding author: Valentina Donzella.*)

Yiting Wang, Boda Li, Zixiang Wei, and Anima Rahman are with WMG, University of Warwick, CV4 7AL Coventry, U.K., and also with the School of Engineering and Materials Science, Queen Mary University of London, E1 4NS London, U.K.

Daniel Gummadi and Haonan Zhao are with WMG, University of Warwick, CV4 7AL Coventry, U.K.

Valentina Donzella is with the School of Engineering and Materials Science, Queen Mary University of London, E1 4NS London, U.K. (e-mail: v.donzella@qmul.ac.uk).

Digital Object Identifier 10.1109/TITS.2025.3600557

¹Society of Automotive Engineers (SAE) categorises autonomous vehicles into six levels: L0 (human driving), L1 (hands-on/shared control; human driving), L2 (hands off; human driving), L3 (human for Dynamic Driving Task fallback; machine driving), L4 (mind off; machine driving in Operational Design Domain) and L5 (machine driving under all conditions).

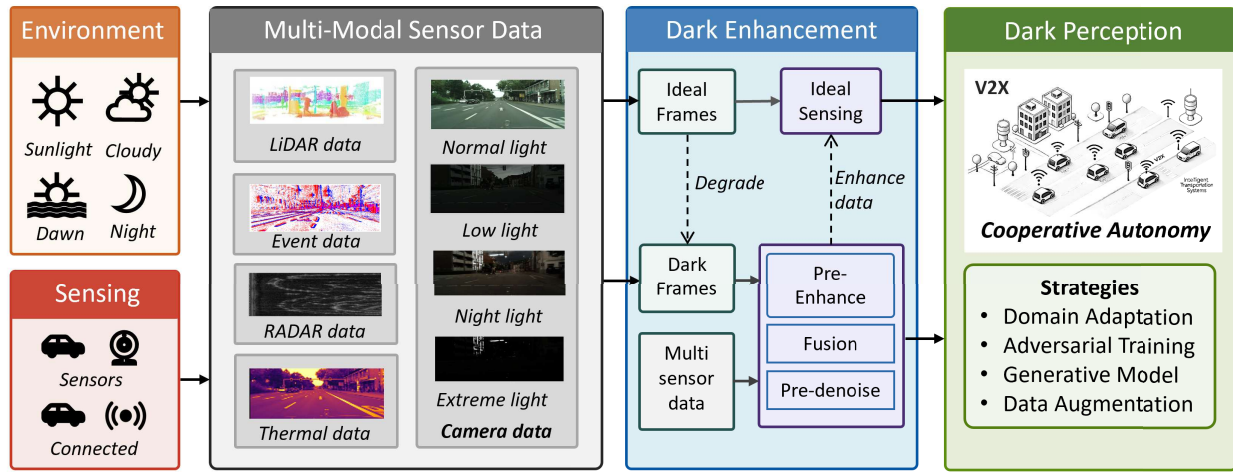


Fig. 1. The high-level pipeline of the DarkSensing. The dotted lines mean the process is optional, as illustrated in different methods. Visualised data is from [8] and [9].

the development of deep learning (DL) and machine learning (ML) techniques and the emergence of new smart sensor technologies (e.g. event cameras, 4D RADAR, new smart night vision sensors), there are new ways and technologies (single or multi-modality) to improve vehicular sensing to allow vehicles to sense in the dark, hereafter named DarkSensing. DarkSensing is essential to ensure the safety and effectiveness of AD at all automation levels. Two approaches are used mainly to improve sensing in the dark: improving sensor data itself (hereafter *dark enhancement*) and improving the performance of perception algorithms that process dark sensor data from single or multiple sources (hereafter *dark perception*).

The intricate nature of DarkSensing relies on a diverse array of supporting technologies crucial for addressing the challenges of dark driving scenarios (Fig. 1). Two fundamental pillars are perception sensor technologies and data-driven dark perception. Perception sensor technologies, including LiDAR, RADAR, and infrared cameras, form the cornerstone of DarkSensing. Sensor data pre-processing involves efficient filtering and interpretation of raw sensor inputs, optimising the quality of information available for subsequent stages. On the other hand, sophisticated data-driven algorithms can be employed for enhancement or perception (e.g. detection, semantic segmentation) in the dark. This enhancement process, integrated into the perception module of the AD pipeline, significantly improves the system's ability to discern crucial details and make robust decisions in challenging lighting conditions. While different algorithms may be used for daytime and nighttime perception tasks, improving nighttime-specific algorithms to address the unique challenges of dark environments remains critical. Vehicle-to-everything (V2X) communication also has great potential to enhance cooperative perception by improving the safety awareness of individual sensors or vehicles under low visibility.

Within the Operational Design Domain (ODD), illumination is inherently limited under dark conditions. For example, the recent PAS 1883:2020 standard defines only two lighting categories (i.e. night light and low ambient light) without sufficiently detailed scenarios or hardware-level explanations of

how different darkness levels impact perception performance. To address this gap, our survey expands the scope to include three distinct darkness conditions: low-light, night light, and extreme light, covering both conventional and deep learning techniques aimed at ensuring the safe operation of AVs under these varying conditions. This comprehensive review supports a more nuanced ODD definition that better accounts for the limitations of current sensing technologies across diverse low-illumination scenarios, thereby enhancing the safety of assisted and automated driving (AAD). In this context, we categorise darkness based on scenario characteristics rather than rigid illuminance thresholds due to significant variations in artificial lighting worldwide and differing definitions in the literature. For instance, according to the PAS 1883:2020 taxonomy, night light is defined as illumination below 1 lux, whereas low-light ranges from 1 to 2000 lux [11]. Other studies define dark light conditions between 0.2 and 5 lux [12], with extreme low-light imaging considered below 1 lux [13], [14]. Some literature also classifies nighttime urban scenes as having illumination greater than 10 lux when captured by auto-exposed cameras [14]. Given these discrepancies, we avoid strict lux-based cutoffs and instead adopt the following scenario-based definitions:

- **Low-light:** a uniform darkening of images, with somewhat visible boundaries of objects. Include scenarios where ambient light is reduced, with some discernible visibility of objects.
- **Night light:** simulates urban areas with traffic lights, headlights, and colourful streetlights that may cause a glare effect. It replicates night in urban areas with various artificial lights.
- **Extreme light:** darker illumination compared to both low-light and night light. Refer to extremely limited light, such as in sparsely populated rural areas or deserts.

Contribution. The main contribution of this survey can be summarised as follows: (I) a systematic survey of the ITS sensing in the dark, including hardware (perception sensors) and software (perception algorithms) perspectives with multi-modal sensors; (II) Qualitative and quantitative case studies

of the current dark synthetic, enhancement, and perception techniques to illustrate limitations and potential future research directions; (III) the creation of an online platform that covers many popular DarkSensing methods and datasets, which can speed up the development of this field.

Survey Structure. The article is organised as follows: Section I introduces the concept and definition of the scope of the DarkSensing survey. Section II gives a comparison of various perception sensor technologies and their hardware performance in the dark. Section III illustrates the current sensor data enhancement methods in the dark. Section IV, Section V and Section VI dive into detection, scene understanding, and cooperative autonomy (e.g. tracking, planning, and cooperative sensing) in the dark. Finally, we discuss case studies, open challenges, and future directions in Section VII and the conclusion in Section VIII.

II. SENSORS IN THE DARK

Robust sensing in dark environments begins with the sensing hardware. Different sensor modalities offer unique strengths and limitations under low-luminance conditions. This section reviews key sensor technologies—such as cameras, LiDARs, RADARs, thermal cameras, and event-based sensors—and analyses their effectiveness, cost, and practicality for DarkSensing applications.

A. Automotive Perception Sensors

Different types of vehicular perception sensors have different capabilities with respect to the noisy automotive environment, and they are described below.

1) *Camera Sensors:* CMOS imaging sensors are widely used in automotive due to their low power consumption, small form factor, and high integration capabilities [10], [15]. However, CMOS sensors are traditionally affected by increased noise in dark conditions. Performance in the dark is mainly hindered by the quantum efficiency of the sensor and read noise [10], [16]. Achieving the necessary sensitivity and high dynamic range in the dark is the key to the safety and robustness of vision-based vehicular sensing in the dark. The output data type from most image sensors is typically in the form of Red, Green, Blue (RGB) data. This RGB output is often heavily processed from the raw data by an Image Signal Processor (ISP) to enhance image quality through operations such as noise reduction, white balance, and tone mapping. Some advanced sensors allow developers to tune the ISP settings, enabling a customised image processing pipeline. In certain cases, it is also possible to access raw sensor data, captured directly from the sensor without any ISP processing (e.g. before demosaicing). This raw data preserves the fullest sensor information which can be particularly useful for low-level computer vision tasks or custom processing needs.

2) *Infrared Cameras:* Infrared cameras do not rely on visible light, so they are immune to performance degradation in the dark. This capability makes them suitable for perception tasks to support nighttime driving [17], [18], [19], [20], [21]. Unlike near-infrared (NIR) or short-wave Infrared (SWIR), long-wave Infrared, also known as thermal infrared sensors,

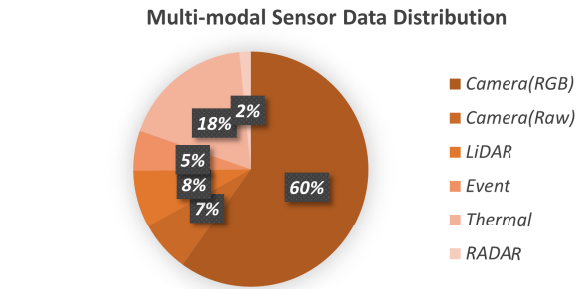


Fig. 2. The distribution of the multi-modal sensors reviewed in this survey. Most commonly used methods are based on cameras.

measures the thermal energy emitted by objects [22]. Thermal cameras exhibit resilience in poor lighting conditions and excel in detecting living creatures, so they are key to supporting the detection of Vulnerable Road User (VRU) at night [23]. Additionally, thermal cameras have demonstrated the ability to withstand flare from oncoming traffic headlamps, a challenging scenario for visible range, NIR and SWIR cameras [24]. Infrared imaging sensors do have limitations, such as reduced contrast and low resolution, sparse spatial details, and blurred edges, primarily due to the longer wavelengths of infrared radiation [25]. Despite these challenges, thermal cameras are increasingly adopted to support AD applications. For example, FLIR provides thermal imaging cameras for Zoox Robotaxi [26], Night Owl Plus Infrared Deer Detector is another commercial application [27]. Similar to visible-range cameras, thermal cameras typically output raw sensor data as a single intensity value per pixel, where each value corresponds to the detected thermal radiation and can be mapped to a temperature. These raw thermal data can be post-processed into greyscale images, where the intensity of the pixels reflects the relative temperature, or into RGB (false-colour) images, where colour gradients are applied via temperature lookup tables to facilitate visual interpretation.

3) *Other Perception Sensors:* Other perception sensors, such as LiDAR and RADAR, are less influenced by the dark due to their “active” operational principles. The varying ambient light has a negligible impact on the transmission and reception of either laser pulse or radio waves, ensuring reliable robustness irrespective of lighting conditions. This makes LiDAR and RADAR reliable assets for maintaining perception capabilities in challenging lighting environments. However, there have been constant efforts to address long-standing limitations on these sensors: new beam-steering and packaging technology in solid-state LiDAR to reduce cost, while 4D RADAR systems use advanced antenna array designs and signal processing techniques to improve spatial resolution. Nonetheless, these active sensors generally do not provide the same rich information on object shapes, edges, and textures as imaging sensors.

Despite these drawbacks, Waymo’s 5th generation truck hardware integrates thermal cameras, conventional cameras, LiDAR and RADAR for improved detection at night and in adverse weather [28]. *Event cameras* are vision sensors designed to mimic certain aspects of biological vision by

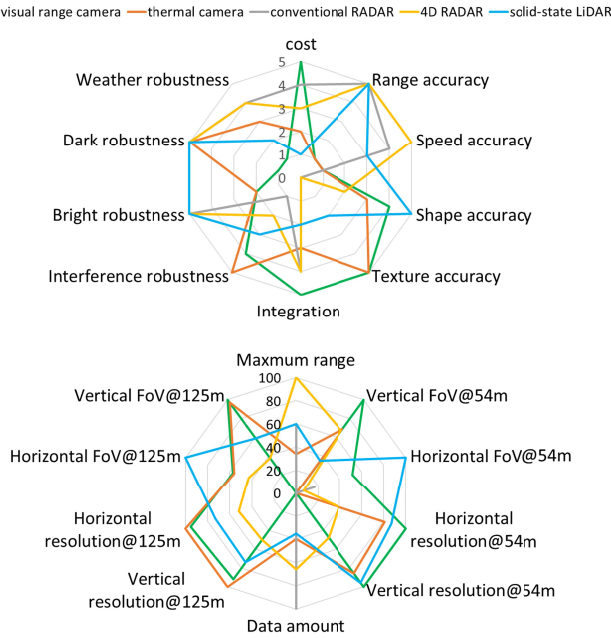


Fig. 3. RADAR plots comparing perception sensors with a focus on illumination-related performance. Comparisons of the five key sensor modalities in DarkSensing applications. (a) at the top: analysis across general performances such as robustness, cost, accuracy, and integration complexity; (b) at the bottom: visualised performance indicators from Tab. II.

asynchronously outputting data only when a pixel detects a change in light intensity, rather than capturing frames at fixed intervals. However, event cameras face challenges in extremely low-light environments where minimal intensity changes occur, often resulting in sparse data and increased noise [29], [30]. Additionally, they tend to be lower in resolution, and are more expensive and bulkier than traditional cameras.

B. Vehicular Sensors Discussion and Comparison

The performance comparisons between the aforementioned perception sensors have been conducted in various studies in different sets of key performance indicators [31], [32], [33], [34], [35]. A RADAR map, focused on the sensor performance in relationship with illumination, is provided in Fig. 3 (a). Ten key performance indicators are selected to account for the focus of this paper, and the score of each indicator is on a scale of 1 to 5, with 1 meaning lower performance and 5 indicating best performance.

Moreover, the comparison of perception sensor performance is carried out by analysing and summarising the information collected from real automotive sensor specifications and datasheets. The same five perception sensors (*i.e.* imaging cameras, thermal cameras, traditional RADAR, 4D RADAR, and solid-state LiDAR) are selected for the RADAR map Fig. 3 (b). This set of sensor performance indicators describes how well a volume of space is covered by the data generated by a sensor. To be comparable and objective, the circumferential angle theorem is used to formulate the FoV and resolution into meters at two cross-sections facing towards the sensor.

One is for the near field FoV and resolution, is at around 54m in front of the sensor, and the other is at 125m. These two distances are sampled according to the 4-sec time-to-collision (TTC) from the European New Car Assessment Programme (NCAP) VRU test protocol for AEB testing and UK speed limits for cars and motorbikes in built-up areas, which is 30 mph and on the motorway, which is 70 mph [36], [37], [38], [39]. To formulate the maximum range of the visual range camera, it is assumed that in a perfect scenario, a dummy target should be covered at least by 10 pixels in height. To formulate the horizontal FoV of a 360-degree LiDAR sensor, the perimeter at 54m or 125m is calculated. According to the RADAR plots, visual range cameras and solid-state LiDARs are the preferred technologies, excelling in various aspects of the comparison. Visual range cameras are the option for providing rich and accurate colour information with low cost and easy integration.

As shown in Fig. 2, cameras are the most commonly used sensors within the methods reviewed in the survey, whereas RADARs are rarely employed. However, cameras have the worst robustness in extreme light conditions because of the limited photons captured. Solid-state LiDARs are the options for improved performance in terms of spatial coverage and data robustness. 4D RADARs are the best choice to offer accurate speed information within a volume of space over long distances. Under extreme dark conditions, the performance of thermal cameras remains unaffected. They achieve top scores in dark-condition robustness together with active sensors such as RADARs and LiDARs. In addition, thermal cameras are the only passive sensing technology in the comparison capable of providing image-like output with spatial detail based on temperature variations, largely unaffected by ambient lighting or mutual interference from the scene. They remain effective even under extreme lighting conditions, such as complete darkness or intense glare, although their resolution and texture detail are generally lower than that of visible-range cameras.

Tab. I compares seven representative datasets that provide sensor data collected under dark sensing conditions. The LOL dataset focuses on indoor low-light enhancement, offering 500 paired images for supervised training. BDD100K is a large-scale driving dataset with diverse conditions, including approximately 27.9k nighttime frames for training. It contains scenes from both urban and rural areas with a good mixture of artificial lighting, such as streetlights and headlights. However, the images are ISP-processed images. Raw data and HDR data are not provided. Oxford RobotCar offers multi-sensor urban raw sensor data with temporal consistency under various weather conditions, including a good amount of noisy data with oncoming traffic headlight interference. However, it does not offer accurate ground truth. ACDC specialises in adverse driving conditions, providing high-quality annotations for fog, night, rain, and snow scenarios with a relatively smaller scale of nighttime driving data. NightCity targets semantic segmentation in complex urban nights, featuring extreme exposure variations [14]. ROD focuses on object detection using HDR raw sensor data, preserving high dynamic range but requiring specialised processing. MUSES provides multi-modal perception data for illumination uncertainty, enabling cross-

TABLE I
COMPARISON OF REPRESENTATIVE DATASETS USED FOR DARK PERCEPTION RESEARCH

	LOL	BDD100k	OxfordROBOT	ACDC	NightCity	ROD	MUSE
							
Application	Image enhancement	Detection segmentation	Localisation and mapping	Semantic segmentation	Semantic segmentation	Object Detection	Panoptic segmentation
Camera	Various cameras	NexarDashboard cameras	Bumblebee [®] XB3 - Sony ICX445 Sensor	1080p GoPro Hero5 camera	A driving recorder	Sony IMX 490 CMOS sensor	TRI023S-C -Lucid Vision
Format	24-bit PNG	24-bit JPEG	Raw 8-bit Bayer Format	24-bit PNG	24-bit PNG	Unrectified 24-bit RAW	24-bit PNG
Resolution	400×600	1280×720	1280×960	1920×1080	1024×512	2880×1856	1920×1080

TABLE II
COMPARISON OF AUTOMOTIVE PERCEPTION SENSOR TECHNOLOGIES CONSIDERING SELECTED SENSOR PERFORMANCE INDICATORS. STATISTICS FROM REPRESENTATIVE COMMERCIAL HARDWARE. EXACT FIGURES DEPEND ON THE SPECIFIC MODEL AND INTEGRATION

Performance indicators	Visual range camera	Thermal camera	Conventional RADAR	4D RADAR	Solid-state LiDAR
Maximum range	150m	200m	250m	350m	250m
Horizontal FoV@54m	179m	90m	113m	99m	339m
Vertical FoV@54m	179m	90m	19m	85m	42m
Horizontal resolution@54m	0.1m	0.28m	12m	1.4m	0.1m
Vertical resolution@54m	0.16m	0.34m	19m	1.4m	0.1m
Horizontal FoV@125m	218m	207m	39m	142m	785m
Vertical FoV@125m	218m	207m	31m	65m	98m
Horizontal resolution@125m	0.12m	0.04m	69m	1.4m	0.22m
Vertical resolution@125m	0.2m	0.06m	31m	1.4m	0.22m
Data rate	3500Mbps	250Mbps	10Mbps	45Mbps	350Mbps

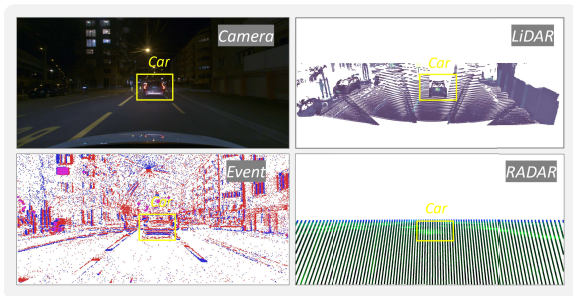


Fig. 4. Synchronised frames from MUSE dataset comparing four sensor modalities of data collected in the dark [9]. The comparison shows the wider LiDAR FOV, and the 'rarely visible car' captured by the camera data can be identified in LiDAR data and the event data.

sensor fusion. The MUSE dataset focuses on multi-modal perception under illumination uncertainty, providing 2,500 high-resolution images synchronised with LiDARs, RADARs, and event cameras. It includes 2D panoptic annotations for handling occlusions and covers challenging lighting conditions such as adverse weather scenarios at night. Its primary strength lies in the ability to fusion and compare between sensors, particularly by using LiDARs for low-visibility regions, as shown in Fig.4.

III. DATA ENHANCEMENT IN THE DARK

This chapter reviews and discusses current dark enhancement methods, with a primary focus on automotive cameras.

Low-light image enhancement (LIE) plays a crucial role in enhancing visibility in dimly lit areas. These methods can serve as a crucial pre-processing step to improve perception performance in downstream tasks (Fig. 5). Based on sensor data formats, we categorised them into vision-based and fusion-guided methods. Tab. III presents current paired datasets of low- and normal-illumination conditions, while Tab. IV gives a summary of representative dark enhancement methods across these modalities, highlighting their strengths and limitations.

A. Vision-Based Enhancement (RGB)

Colour images have been traditionally used for LIE, where several deep-learning methods are based on this data format. Histogram Equalisation (HE) [40] and Retinex theory [41] are the commonly used techniques for the traditional RGB enhancement approaches. Retinex-based methods enhance low-light images by decomposing them into reflectance and illumination maps, while HE-based techniques remap intensity levels based on distribution. The naive implementation of these techniques for RGB enhancement is mainly for tasks like vehicle detection, and driver face detection [42], [43], [44]. However, these methods often exhibit magnified noise and lack adaptability to different image regions. To solve these problems, some more recent work implements new techniques such as dehazing or HSV (hue, saturation, and value) data format transformation to prevent over-enhancement

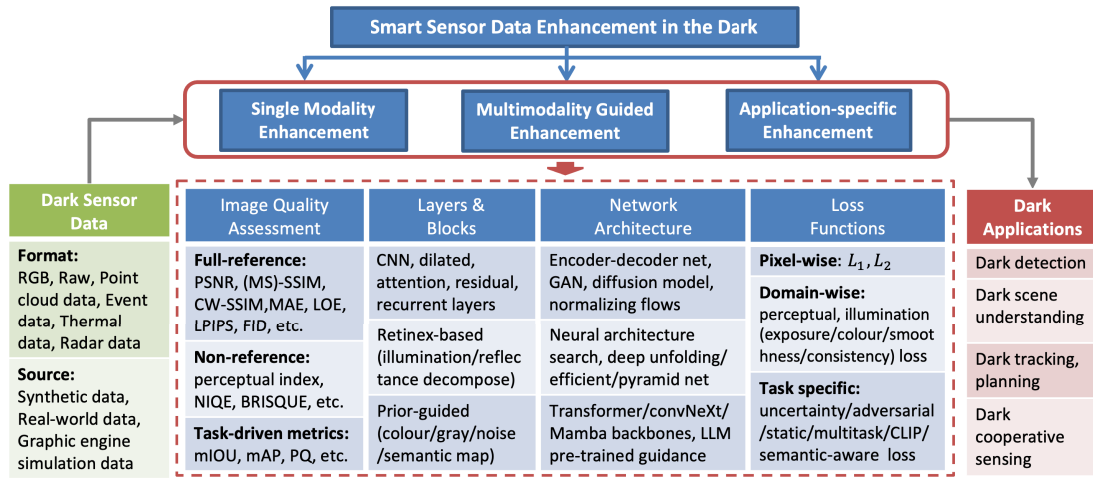


Fig. 5. Visual representation of different types of smart sensor data enhancement in the dark. It includes the Dark Sensor Data Format, single or multi-modality enhancement, common metrics, architecture, loss functions used and dark application-specific tasks.

TABLE III
SUMMARY OF DARK-PAIRED DRIVING DATASETS
FOR VISION ENHANCEMENT

Dataset	Format	Videos	Scenario	Scale
SID [12]	Raw	✓	outdoor/indoor	5.1k
LOL [47]	RGB	✗	mostly indoor	500
VE-LOL [48]	RGB	✗	outdoor/indoor	2.5k
SCIE [49]	RGB	✗	outdoor/indoor	4.4k
DRV [50]	Raw	✓	outdoor/indoor	202
SMOID [51]	Raw	✓	outdoor	179
SDSD [52]	Raw	✓	outdoor/indoor	150
Flare7K [53]	RGB	✗	outdoor	7k
MCR [54]	Raw	✓	outdoor/indoor	498
RLED [30]	Event	✗	mostly outdoor	64.2k
SDE [55]	Event	✗	mostly indoor	30k

and high-saturation problems in traffic monitoring systems [45], [46].

With advancements in deep learning (DL) techniques, numerous RGB enhancement methods have been developed to address visibility issues [56]. State-of-the-art LIE methods leverage convolutional neural networks, transformer models, and diffusion networks for supervised and unsupervised learning [57], [58], [59], [60], [61], [62], [63], [64]. The ultimate goal of these methods is to improve the similarity of the enhanced images with their reference normal light images. There is less work employing DL-based enhancement for supervised dark enhancement in AAD, due to the difficulty in capturing the paired driving datasets. Some methods are implementing paired multi-exposure images with static objects via lightweight networks [65], [66], [67]. Qin et al. proposed a multi-task learning method for enhancement with detection, using both the high-frequency and low-frequency information for training [68]. Prompt-guided interactive learning has also been proposed for backlit image enhancement in a recent research [64].

To further handle the effect of strong lights in low illumination, Jin et al. introduced an unsupervised nighttime enhancement method that suppresses light impact in

bright areas while enhancing intensity in dark areas, utilising structure and high-frequency information [69]. Additionally, Dai et al. developed a transformer-based image restoration structure for removing different types of flare from nighttime images, achieving state-of-the-art performance [53]. Another group of research focuses on using adversarial networks (*e.g.* GAN) to synthesise paired daytime and nighttime driving images for network retraining and image enhancement [8], [70], [71], [72], [73]. For example, LE-Net synthesises nighttime images from daytime images using the conventional gamma transform and contrast adjustment method for training the enhancement model [70]. Wang. et. al generated low-light, night light and extreme-light images using the combination of the NN and CycleGAN [8], [73]. These methods can generate visually pleasant images that are similar to the different dark driving scenarios.

B. Vision-Based Enhancement (Raw)

Raw images are increasingly prevalent in automotive applications as they contain the pristine collected information. To convert these raw images into RGB format, they typically go through an ISP pipeline, which might include noise reduction, exposure, adjustment, contrast enhancement, and white balance correction [12]. However, the limited information captured in low-light raw images often leads to suboptimal RGB output when processed by conventional ISP methods. To address this limitation, recent approaches have focused on redesigning the ISP process specifically for low-light conditions, aiming to enhance raw images before they are converted to RGB. Notable examples include techniques such as the See-in-the-Dark (SID) method [12], the de-Bayer-filter approach [54], and other adaptive strategies [74]. Zamir et al. tackled the issue of over-smoothing in low-light raw image enhancement by introducing an end-to-end framework designed to preserve details while enhancing image quality [13]. In a related effort, Morawski et al. proposed a novel ISP pipeline with improved generality across various camera sensors, specifically

TABLE IV

A SUMMARY OF REPRESENTATIVE DARK PERCEPTION ENHANCEMENT METHODS ACROSS DIFFERENT SENSOR DATA FORMATS. METHODS ARE CATEGORIZED BY INPUT MODALITY, WITH REPRESENTATIVE WORKS AND THEIR KEY STRENGTHS (A) AND LIMITATIONS (D)

Sensor Data	Representative Methods	Advantages (A) VS disadvantages (D)
Camera-RGB	Conventional methods [42]–[45], [78], [46], [77]	A: Simple, interpretable, low computation cost. D: Limited generalization to real-world dark conditions, susceptible to noise/artifacts.
Camera-RGB	Deep Learning-based [57]–[68], [70], [71], [83]–[85]	A: Learnable representations, strong enhancement quality, good generalization ability. D: Require large labelled datasets, sensitive to domain shift, less interpretable.
Camera-Raw	ISP-involved [12], [13], [51], [54], [75], [76], [86], [87]	A: Bypasses ISP pipeline; wider color gamut, less noise, better signal recovery. D: Requires raw data access, complex training, higher computational cost.
Thermal-Infrad	Thermal-guided methods [82]	A: Not affected by light, robust in complete darkness. D: Low spatial resolution, often lacks texture detail.
Camera-Event	Event-guided methods [52], [88], [89]	A: High dynamic range, low latency, blur-free in motion. D: Requires specialised models, limited spatial resolution, challenging integration.

tailored for low-light conditions [75]. Additionally, Lamba and Mitra developed a simple amplification module aimed at pre-enhancement, enabling real-time detection in extremely dark environments [76]. The enhancement of dark raw video has also been explored, with Chen et al. introducing a 3D U-net architecture along with a new benchmark for dark video enhancement [50].

C. Fusion-Based Enhancement (Thermal)

In dark environments, enhancing images through the fusion of thermal and visible (RGB) data has gained attention due to the inherent limitations of image quality in such conditions. Supervised thermal-guided enhancement typically involves the fusion of thermal images with high-resolution RGB data. These methods aim to improve the brightness, contrast, and colour fidelity of RGB images by incorporating complementary information from thermal imagery. For instance, Zhou et al. [77] developed a technique that uses a filtering module to denoise visible images before fusion. Similarly, Yang et al. [78] applied Retinex theory to image decomposition, enhancing both high- and low-frequency details during the fusion process. In addition to these techniques, significant work has been done on infrared and visible image fusion using novel network architectures. For example, TGFuse introduced a transformer-based fusion module with adversarial learning to enhance fused images [79], while LRRNet proposed a representation-learning-guided fusion network [80]. These fusion-based enhancement methods are particularly valuable for improving AD target recognition in low-light or dark environments, where conventional RGB imaging alone is insufficient. Furthermore, some thermal super-resolution (SR) methods have been proposed to improve the interpretability and usability of thermal images by aligning them more closely with visible spectrum data using Generative Adversarial Networks (GANs) [81], [82]. These methods are potentially another way of improving the content of dark images before the fusion process. These enhancement-based methods can play a pivotal role in enhancing the robustness of AD target recognition algorithms in dark environments.

IV. DETECTION IN THE DARK

Object detection in AAD involves the ability to accurately and efficiently understand the location and size of the object (e.g. cars, pedestrians, and road signs) for safety purposes.

This subsection discusses existing work for object detection under dark conditions (Tab. V).

A. Vision-Based Detection (RGB)

Several detection methods are designed for full-colour images or video frames using DL methods. Arora and Kumar categorise vehicle detection methods into knowledge-based, motion-based, and vision-based methods for day and night [114]. Their experiment indicates that the conventional hand-crafted features, such as Haar-like features, fail to represent the diverse vehicles and other objects of different shapes [114]. Most of the current dark detection methods, therefore, are DL-based methods, based on popular one-stage and two-stage detectors (e.g. YOLO, Faster-RCNN)

These methods often involve adding a pre-processing module to enhance the dark images or enhance the degraded feature [90], [91], [96], [99]. For example, DarkYOLO proposed a glue layer and a generative model to combine the pre-trained models (i.e. SID and YOLO) [91]. Darknet is used for Automatic License Plate Recognition [90]. PE-YOLO proposes an end-to-end framework with a pyramid enhancement network before the detector [96]. WSA-YOLO proposed an adaptive residual feature block and parameter prediction block for decomposed dark images to improve detection [99].

Another group of solutions is based on domain adaptation methods focused on reducing the domain gap between the normal light and dark light images [92], [98]. For example, Hla-face proposes a dark face detection framework [92]. Specifically, the bidirectional distortion (illumination, noise, and colour) enhancement module and multi-task cross-domain self-supervised learning module are designed for low-level and high-level adaptation [92]. 2PCNet is proposed with a two-phase learning strategy, where the higher confidence prediction from the teacher model is utilised for student model learning in the first phase. In the second phase, the teacher model re-evaluates results from the student model to improve detection accuracy via the teacher-student learning [98].

In addition, multi-task learning is also used for detection and enhancement. For example, DE-YOLO designs a lightweight joint enhancement and detection model for detection-driven enhancement, where the global information from the low-frequency is used to guide the high-frequency detail reconstruction [68]. Except for the current real-world

TABLE V

COMPARISON OF REPRESENTATIVE METHODS FOR OBJECT DETECTION IN DARK ENVIRONMENTS. COLUMNS INCLUDE PUBLICATION DETAILS, USE OF GROUND TRUTH (GT) LABELS FOR DARK IMAGES, INPUT DATA MODALITIES, DETECTOR ARCHITECTURES, DATASETS USED, AND KEY METHODOLOGICAL HIGHLIGHTS. METRICS INDICATE THE DETECTION SCORE ON THE SPECIFIC DATASETS. (N) DENOTES RESULTS OBTAINED FROM THE NIGHTTIME SUBSET OF THE DATASET

Paper	GT	Modality	Detectors	Dataset	Metrics	Highlight
Darknet [90] IVC'19	✓	RGB	YOLO	AOLP dataset	A(mAP):0.83	The first algorithm to detect and recognise characters and digits in license plate
DarkYOLO [91] ECCV'20	✓	RGB Raw	YOLO	SID	S(mAP):0.56	A glue layer and generative layer to combine pre-trained SID and YOLO models
Hla-face [92] CVPR'21	✗	RGB	DSFD	DARK FACE	D(mAP):0.44	A joint low-level and high-level adaptation framework for dark face detection
Li et.al. [93] ICCV'21	✓	RGB Raw	Faster- RCNN	Synthetic	S(mAP):0.66	Space-time non-local module and knowledge distillation for photon-limited images
NighttoDay [94] TIV'21	✗	RGB	Any	BDD 100k	B(F1):0.75	Online night-to-day image-to-image translation for re-training detectors
MAET [95] ICCV'21	✓	RGB Raw	YOLO	ExDark	E(mAP):0.56	Consider the physics of sensors and ISP-process with multi-task learning
GenISP [75] CVPRw 2022	✓	RGB Raw	Any	NOD	N(mAP):0.25	A neural ISP process to improve generability and enhance the raw data directly
DE-YOLO [68] ACCV'22	✓	RGB Raw	YOLO	ExDark	E(mAP):0.52	A lightweight joint enhancement and detection model with frequency-based detail restoration
PE-YOLO [96] ICANN'23	✓	RGB	YOLO	ExDark	E(mAP):0.78	A pyramid enhancement module for end2end dark object detection
DarkFeat [97] AAAI'23	✓	Raw	MulDet	MID	/	A noise-robust detector with a high fidelity ISP pipeline for extremely low-light
2PCNet [98] CVPR'23	✗	RGB	Faster- RCNN	BDD100k SHIFT	B(mAP):0.46	A two-phase consistency unsupervised day-to-night domain adaptation network
WSA-YOLO TIM'24 [99]	✓	RGB	YOLO	ExDark	E(mAP):0.84	Enhance the decomposed reflectance and illumination separately for detection
MBNet [100] ECCV'20	✓	RGB Thermal	SSD	KAIST CVC	/	Adaptively align and balance RGBT features based on the illumination conditions
LLVIP [101] ICCV'21	✓	RGB Thermal	YOLO	KAIST LLVIP	L(mAP _{rgb}):0.50 L(mAP _{ther}):0.62	a visible-infrared paired dataset for low-light vision at very dark scenes
Zhou et al. [102] Remote Sens.'21	✓	RGB Thermal	RetinaNet	FLIR KAIST	F(mAP):0.73	Temperature characteristics of thermal images and illumination-aware attention for fusion
TIRNet [103] Applied Intel.'21	✓	Thermal	SSD	KAIST CTIR	C(mAP):0.75	A lightweight SSD-improved residual module in thermal pedestrian detection
GAFF [104] WACV'21	✓	RGB Thermal	RetinaNet	KAIST FLIR	/	Combining intra- and inter-modality attention to achieve adaptive weighing
DAYOLO [105] TIV'22	✓	RGB Thermal	YOLO	KAIST BDD100k	K(AP _{people}):0.6 B(mAP _n):0.35	A domain adaptation method with auxiliary domain and category-consistency regularisation
LSNet [106] TIP'23	✓	RGB Thermal	MobileNet	VT5000	V(MAE):0.036	A lightweight spatial-guided network with geometric and semantic transfer learning
YOLO-CIR [107] Infra. Phys.'23	✓	Thermal	YOLO	FLIR KAIST	/	Detect on the 14-bit raw thermal images directly using improved ConvNeXt
LRAF-Net [108] TNNLS'23	✓	RGB Thermal	YOLO	FLIR,LLVIP VEDAI	L(mAP):0.66 V(mAP):0.59	A feature-enhanced long-range attention fusion network with data augmentation
ICAFusion [109] Pattern Reco.'24	✓	RGB Thermal	YOLO	Kaist,FLIR VEDAI	F(mAP):0.41 V(mAP):0.45	A dual cross-attention transformer for effect feature fusion
TransFusion CVPR'22 [110]	✓	RGB LiDAR	CenterNet VoxelNet	nuScenes Waymo	N(mAP):0.69 W(mAP):0.66	a robust fusion solution with a soft-association mechanism to handle inferior image conditions
Mutual Fusion TCE'24 [111]	✓	RGB LiDAR	ResNet-50 VoxelNet	nuScenes Waymo	N(mAP):0.70 W(mAP):0.68	Learn better semantic spatial information, avoid negative transfer with fused features
RENet [112] ICRA'23	✓	RGB Event	MOD	DSEC-MOD	D(mAP ₅₀ ⁿ):0.38	A moving object detection method with temporal aggregation and bi-directional fusion
EOLO [113] ICRA'24	✓	RGB Event	YOLO	PASCAL MSCOCO	P(AP ₅₀ ⁿ):0.61 M(AP ₅₀ ⁿ):0.35	An all-day object detection framework with symmetric RGB-event fusion

captured datasets, lots of generative methods are considered for improving the dark perception directly with retraining. For example, some methods change the existing labelled daytime detection dataset into nighttime with UNIT [94], [115], [116], GAN [72] and diffusion [117].

B. Vision-Based Detection (Raw)

Directly detecting objects using raw images has been highly explored, as raw images contain more pristine information and have the potential to adaptively tune the ISP process suited for dark detection. For example, Li et al. enhance

the Faster-RCNN with knowledge distillation to improve the feature representation ability of the images under extreme photon-limited scenarios [93]. As the current methods tend to use a fixed ISP process, MAET proposed a multi-task-learning strategy for intrinsic physics models-based ISP tuned for dark detection on the raw data [95]. Similarly, GenISP tried to solve the limited generalisation ability caused by the different spectral sensitivity of the various sensors by proposing a neural ISP process to enhance the nighttime detection performance [75]. DarkFeat proposed a noise-robust detector with a high-fidelity ISP pipeline for extremely low-light image detection [97]. These methods are specially designed for carefully captured datasets via specifically tuned sensors; it remains challenging to answer the question of which detection strategy is better for automotive sensor applications with more complex scenarios and limited computational resources and communication bandwidth.

C. Fusion-Based Detection (Thermal)

Methods have been explored to detect pedestrians purely using the 14-bit raw thermal data or the 8-bit processed data due to their robustness towards illumination [103], [107]. However, as the thermal images present lower resolution and limited structure details, more methods use visible-thermal fusion in object detection to leverage the advantages of both sensor types. Various strategies, such as adaptive feature alignment, prior information (*i.e.* temperature masks and boundary masks), and attention guidance (*e.g.* intra- and inter-modality attention), enhanced features with long-range ability are used during the feature fusion process to achieve better detection accuracy [100], [102], [103], [108], [109]. The main characteristics of each method are summarised in Tab. V.

D. Fusion-Based Detection (LiDAR)

Recently, LiDAR point clouds have also been explored to support vision in the dark, given LiDAR's robustness towards variable illumination. For example, TransFusion proposed the first robust camera-LiDAR fusion-based detection network. After initial detection from LiDAR point cloud data, contextual information is adaptively fused into the second layer from the camera data to achieve robust detection with spatial and contextual information [110]. MutualFusion [111] detects in parallel camera data and LiDAR data with fused semantic-spatial information, where the semantic information is transferred from the enhanced camera and the spatial information is from the LiDAR point cloud. All of these methods involve adding the camera detection results to the initial LiDAR detection results to amend the sparse semantic information from the LiDAR point cloud.

E. Limitations of the Current Dark Detection Methods

Despite significant progress, existing dark detection methods continue to face key limitations in generalisation, data requirements, efficiency, sensor usage, and architectural design. Many detection models are trained and evaluated on specific datasets captured under constrained or predefined conditions [68], [96], [99]. As a result, they often struggle to

generalise to real-world nighttime environments, where illumination levels, occlusion, and weather can vary significantly, and the enhancement fails. Deep learning-based detectors, especially those that rely on RGB or raw input, typically require large amounts of annotated nighttime data, which is expensive and labour-intensive to acquire. Although domain adaptation techniques might offer partial improvement, they may not solve domain gap problems in the real world.

Architecturally, many earlier methods follow a two-stage pipeline, where image enhancement is firstly applied independently, followed by detection, using pre-trained detectors such as YOLO [91], [96]. While straightforward, this separation lacks joint optimisation and often leads to feature mismatch between the enhancement and detection modules. In contrast, more recent approaches such as DE-YOLO [68] and WSA-YOLO [99] attempt to unify enhancement and detection into an end-to-end framework. These models achieve better feature-level robustness under low-light conditions, but require sophisticated training strategies and remain sensitive to hyperparameter tuning on specific datasets. Raw-based detection methods, such as MAET [95], GenISP [75], and DarkFeat [97], seek to bypass traditional ISP pipelines and directly leverage the richer information embedded in raw sensor data. While promising in terms of dark image recovery, these methods introduce added complexity in ISP modelling and sensor-specific tuning, and they may not meet real-time requirements, particularly in resource-constrained automotive environments.

Domain adaptation and teacher-student frameworks (*e.g.* HLA-Face [92], 2PCNet [98]) tackle domain shift challenges by leveraging unlabelled or weakly labelled data. However, these methods often require multi-stage training and error propagation during pseudo-label generation. Similarly, multimodal fusion-based pipelines, including TransFusion [110] and LRAF-Net [108], integrate RGB with LiDAR or thermal inputs to enhance detection robustness. Yet, they suffer from synchronisation issues, higher latency, and increased model complexity, which again limit real-world deployability.

Overall, while the field has seen promising architectural and algorithmic innovations, existing dark detection methods still face substantial trade-offs between accuracy, generalisability, computational efficiency, and hardware availability. Further work is needed to develop lightweight, adaptable, and sensor-agnostic models for robust deployment in intelligent transportation systems.

V. SCENE UNDERSTANDING IN THE DARK

Scene understanding involves the ability of the AAD system to perceive and comprehend the surrounding environment (*e.g.* cars, pedestrians, and roads). Common techniques include semantic segmentation, instance segmentation, and panoptic segmentation, providing an understanding of the control at a pixel level. This section summarises some of the existing methods with different strategies for scene understanding under dark conditions, Tab. VI.

A. UDA Segmentation Strategy

After reviewing more than 25 relevant methods, it is found that more than 90% of these methods are

TABLE VI

COMPARISON OF METHODS FOR NIGHTTIME SCENE UNDERSTANDING, INCLUDING PUBLICATION DETAILS, WHETHER UNSUPERVISED DOMAIN ADAPTATION (UDA) IS APPLIED, TASKS ABBREVIATED AS SS: SEMANTIC SEGMENTATION, IS: INSTANCE SEGMENTATION, PS: PANOPTIC SEGMENTATION AND DATASET USAGE ABBREVIATED AS DZ: DARKZURICH, BDD: BDD100K, ND: NIGHTTIME DRIVING, ACDC, NC: NIGHTCITY, FT: FREIBURG THERMAL, SYN: SYNTHIA. METRICS INDICATE THE mIOU/AP/PQ SCORE BASED ON THE TASK, E.G. D(mIOU):42.0 MEANS THE BEST mIOU SCORE IS 42 IN THE DZ TEST DATASET

Paper (Year)	UDA	Task	Modality	Dataset	Metrics	Loss	Highlights
GCMA [124] ICCV'19	✓	SS	RGB	CS,DZ ND	D(mIOU):42.0 N(mIOU):45.6	L_{ce}	Curriculum transfer learning to adapt features gradually from day to night
CDAda [130] ICCV'21	✓	SS	RGB	DZ,ND BDD	D(mIOU):45.0 N(mIOU):50.9	$L_{KL}, L_{ce},$ L_{exp}	Day2night inter-domain features and easy-hard split intra-domain features
Dannet [118] CVPR'21	✓	SS	RGB	DZ,ND	D(mIOU):45.2 N(mIOU):47.7	$L_{illu}, L_{fcl},$ $L_{seg}^{static}, L_{adv}$	One-stage adaptation with re-lighted features and day-night discriminators
LIIS [122] TM'21	✗	IS	RGB	LIS	L(AP):34.5	L_{DM}	Wavelet-based fusion on enhanced and denoised low-light image feature
CCDistill [131] CVPR'22	✓	SS	RGB	DZ,BDD ACDC	D(mIOU):47.5 A(mIOU):37.7	L_{seg}, L_{cdc} L_{cfs}	Cross-domain distillation for both content and style features
SFNet-N [119] TITS'22	✓	SS	RGB	DZ,ND CARLA	D(mIOU):56.9 N(mIOU):57.4	L_{illu}, L_{seg}	Multi-scale aligned features from light enhanced carla and transferred images
Similarity minmax ICCV'23 [121]	✗	SS	RGB	DZ,ND	D(mIOU):40.2 N(mIOU):44.9	L_{illu}, L_{sim}	Zero-shot min-max learning from both the image and model levels
Refign [126] WACV'23	✓	SS	RGB	DZ ACDC	D(mIOU):56.2	$L_{align},$ L_{Huber}	Uncertainty-aware paired images alignment and probabilistic label refinement
SePiCo [132] TPAMI'23	✓	SS	RGB	DZ, SYN GTA	D(mIOU):45.4	$L_{cl}, L_{ce},$ L_{reg}	Semantic-guided contrastive learning emphasizing under-represented classes
DIAL-Filters TCSVT'23 [120]	✓	SS	RGB	DZ, ND NC,ACDC	D(mIOU):46.0 ND(mIOU):50.3	L_{seg}, L_{adv} L_{static}	Adaptive filters for double enhancement before and after segmentation
LoopDA [128] WACV'23	✓	SS	RGB	DZ, ND	D(mIOU):50.6 N(mIOU):54.0	L_{ce}, L_1 L_{adv}, L_{perc}	Network training with inner and outer loops for feature learning
DIGA [137] CVPR'23	✓	SS	RGB	BDD	B(mIOU):35.8	/	Backward-free instance-guided test time domain adaptive feature learning
MIC [135] CVPR'23	✓	SS	RGB	DZ ACDC	D(mIOU):60.2 A(mIOU):70.4	L_{ce}, L_{adv} L_{mic}	Masked image consistency module to enhance the spatial correlation feature
CISS [143] TPAMI'25	✓	SS	RGB	DZ,BDD ACDC	D(mIOU):60.7 A(mIOU):69.6	L_{inv}, L_{ce}	condition-agnostic representations by feature invariance loss
LIS [138] IJCV'23	✗	IS	RGB Raw	LIS	L(AP):36.8	L_{is}, L_{ds}	Deep and shallow features enhancement directly on raw data
DSOR [129] IROS'24	✓	SS	RGB	DZ, ND ACDC	D(mIOU):65.2 N(mIOU):58.4	L_{ce}, L_{cl} L_{mix}	Refines at the label and features levels for dynamic and small objects
NightFormer [144] ECCV'24	✗	SS	RGB	CS, NC BDD	N(mIOU):63.5 B(mIOU):38.2	L_{ce} L_{dice}	Amplified attention from texture enhancement and object-level matching
Heatnet [134] IROS'20	✓	SS	RGB Thermal	MF FT	M(mIOU):53.0 F(mIOU):64.9	L_{adv}, L_{ce}	RGBT feature knowledge distillation, with a domain discriminator
ECGFNet [133] TITS'23	✓	SS	RGB Thermal	MF	M(mIOU):55.3	L_{ce}	Embedded control gate block to filter out the most valuable fused features
MTANet [139] TIV'23	✓	SS	RGB Thermal	MF PST900	M(mIOU):56.1 P(mIOU):78.6	L_{ce}, L_{mtl}	A hierarchical feature fusion module with priors (<i>i.e.</i> edge) supervision
Night-TTA [136] TAI'23	✗	SS	RGB Thermal	MF KAIST	M(mIOU):63.3 K(mIOU):47.8	$L_{EN}^{TTA}, L_{KL}^{TTA}$	Singular and fused modality prediction and class-aware TTA
SGFNet [140] TCSVT'23	✗	SS	RGB Thermal	MF PST900	M(mIOU):58.4 P(mIOU):82.8	L_{ce}, L_{Lovas}	Semantic-guided features in encoder and edge features in decoder
CAINet [145] TM'24	✗	SS	RGB Thermal	MF PST900	M(mIOU):58.6 P(mIOU):84.7	L_{Att}, L_{seg}	Complementary global and detailed features for better fusion
Cmda [141] ICCV'23	✓	SS	RGB Event	DSEC DZ	DS(mIOU):60.1 DZ(mIOU):54.8	$L_{ce}, L_{s/t}$	Motion and content feature extractor in cross-modalities and domains
Fusemodnet [142] ICCVW'19	✗	SS	RGB LiDAR	Synthetic	S(mIOU):75.3	L_2, L_{ce}	Data content features and motion flow feature fusion for enhancement
CAFuser [146] RAL'25	✗	PS	RGB LiDAR RADAR Event	MUSES Deliver	M(PQ):78.2	L_2, L_{ce}	A conditional fusion strategy for night adverse weather segmentation

based on unsupervised domain adaptation (UDA). This is mainly due to fewer segmentation datasets for nighttime segmentation. UDA focuses on adapting a model trained in a source domain to perform well in the

target domain, where labelled data may be scarce or unavailable.

Current UDA-based segmentation models utilise different strategies to reduce the domain gap, which can be grouped

into three categories. The first is preprocessing at the image or feature level, to reduce the domain gap. At the feature level, feature alignment is used to reduce gaps caused by low illumination or noise [118], [119], [120], [121], [122]. At the image level, GAN-based image-to-image translation, or transfer learning, is used to translate the labelled daytime images into the nighttime to retrain the model [116], [123]. GCMA and MGCMA proposed a curriculum learning framework that adds an intermediate domain between the day and nighttime (*i.e.* twilight) to progressively take full advantage of labelled synthetic images and unlabelled real images [124], [125].

The second is to leverage coarsely aligned day-night image pairs (*i.e.* captured in the same GPS location) to reduce the domain gap [118], [126], [127], [128]. These methods can improve the quality of the pseudo labels, for example, by the additional supervision of the static background, which is supposed to be the same during both day and nighttime at the same location. However, it also has some limitations, for example, the blending of the dynamic objects may be mixed with the background, leading to the wrong supervision. Secondly, misalignment still exists between day and night images (*i.e.* location and angle changes), worsening the performance. The third group of the UDA-based methods considers not only the inter-domain gaps (*i.e.* different times of the day) but also the intra-domain gaps, such as the unbalanced classes or different segmentation difficulties within the same domain [129], [130], [131], [132].

B. Loss Function Designs

The most commonly used loss functions in segmentation are summarised in Fig. 5 and Tab. VI. Standard loss functions such as cross-entropy loss (L_{ce}) and segmentation loss (L_{seg}) are widely applied in all scenarios where pixel-level classification into multiple classes is required [124], [133]. For dark segmentation, both common domain adaptation loss and dark scenarios, specifically designed novel loss functions have been developed to address the unique challenges posed by dark conditions.

1) *Domain Adaptation Loss*: Perceptual loss (L_{perc}) is designed to reduce the feature discrepancy between domains, enabling better generalisation [128]. Illumination loss (L_{illu}) is employed in conjunction with exposure loss and total variance loss to relight dark images while preserving smoothness and naturalness [118], [119]. Adversarial loss (L_{adv}) is frequently used to remain indistinguishable between the source and target domains [134]. Contrastive loss (L_{cl}) has been used to effectively distinguish semantic class distributions, enhancing inter-class separation in segmentation tasks [132]. Masked consistency loss (L_{mic}) addresses spatial context relationships by focusing on alignment between target images and their masked counterparts, thereby improving spatial awareness [135].

2) *Dark Task-Specific Loss*: Other task-specific loss designs have also emerged to address unique challenges. For example, patch-wise loss maintains local consistency, making it adaptable to straightforward and challenging segmentation subsets [130]. Test-time adaptation loss (L_{TTA}) has been introduced to handle the continuous domain shifts caused

by changing weather conditions, ensuring robust performance under varying environments [136], [137]. Static segmentation loss (L_{static}) has been proposed for pixel-wise supervision of static categories, such as roads or skies, particularly when datasets are collected at the same location. However, its effectiveness can be limited in cases where dynamic objects, such as vehicles or pedestrians, occlude static areas, leading to inaccurate guidance [118], [126]. For instance, segmentation in the dark, instance loss (L_{is}) has been proposed, incorporating components such as classification loss, bounding box regression loss, and per-pixel segmentation loss to handle the specific requirements of this task [122], [138].

C. Multimodal Fusion Strategy

The fusion of multi-modal sensor data has also been largely explored. The mainstream of work lies in the use of visible-thermal fusion for better segmentation performance in the night driving scenes [133], [134], [139], [140]. They aim to leverage the advantages between the two modalities, extracting more detailed and accurate structural features and filtering out the noise. Event-based camera data is proven to have the advantage of capturing better edge information compared to conventional cameras, even under low-light conditions, and is thus considered for fusion with dark visible images for better segmentation performance [29], [30], [141]. In addition, Fusemodnet proposed a supervised RGB and LiDAR feature alignment and fusion method for scene understanding [142].

VI. COOPERATIVE AUTONOMY

In addition to sensor-level and perception-level advancements, dark perception also intersects with broader cooperative autonomy strategies in intelligent transportation systems. Tasks such as depth estimation, object tracking, and path planning become more challenging under low-light conditions due to reduced visibility and data uncertainty. To address these challenges, cooperative sensing via vehicle-to-everything (V2X) communication offers promising opportunities by allowing vehicles and infrastructure to share critical environmental information. We review these emerging technologies and discuss their potential to improve resilience and robustness in dark environments in Section VI.

A. Estimation, Tracking, and Planning

1) *Dark Depth Estimation*: Depth estimation is the process of determining the distance to an object captured in a camera frame, which is crucial in ITSs for accurate obstacle detection, navigation, and safe manoeuvring [147]. Chang and Nguyen addressed the challenge of depth estimation in low-light conditions with a DL-based pre-enhancement embedding defogging to address the difficulty in detecting pedestrians and guardrails [148]. Vankadari et al. proposed a UDA-based monocular depth estimation method with PatchGAN-based adversarial discriminative learning, producing good results on the Oxford night driving dataset [149]. Taking a different approach, Shi et al. tackle monocular depth estimation in challenging nighttime conditions by leveraging both advantages of the RGB and event-based modalities [150].

2) *Dark Tracking*: Simply estimating object location is not sufficient for AAD systems; evaluating dynamic object trajectory and velocity is also essential for accurate motion prediction [151]. Tracking in dark conditions poses additional challenges due to reduced visibility, which can impact the accuracy of trajectory and velocity estimation. Jin et al. propose a method that enhances low-light images (*i.e.* Fusemodnet) with Retinex technology to boost contrast, then refines results with segmentation and bounding box optimisation [152]. Fusemodnet tries to solve real-time moving object detection in the dark, enhancing visible data using point clouds [142]. In this work, both the colour and depth information can be captured in the dark, with optical flow masks to guide the detection of the moving objects. Thermal cameras are also considered and have been proven effective for vehicle detection and tracking at twilight and night. Tian et al. propose the first feature-based online learning method for vehicle tracking [153]. El Ahmar et al. experimentally demonstrate that models trained on thermal images outperformed those trained on visible images under poor lighting conditions, highlighting the advantages of cross-modal learning for robust tracking [154].

3) *Dark Planning*: Path planning involves determining a safe trajectory for the vehicle, an important stage for navigating under various environmental challenges, including dark conditions [151]. There has been limited research on this topic. Recently, a new real-world autonomous driving planning benchmark has been proposed with labels that exclude night data due to its impact on the quality of the perception system [155]. Another newly emerged AAD end-to-end strategy tried to integrate perception to control seamlessly via deep learning [156]. Wang et al. explore utilising the multimodal foundational models to improve the robustness of the AAD systems [157]. Still, there is limited discussion and systematic analysis of the nighttime conditions.

B. Cooperative Sensing

Cooperative sensing leverages shared sensor data through V2X communications to enhance situational awareness, safety, and coordination in multi-vehicle environments [183], [184]. V2X encompasses various communication types, including vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and vehicle-to-network (V2N), all of which have the potential to support collective perception and decision-making when individual vehicle sensors face limitations due to darkness.

1) *Shadow Perception and Infrastructure-Aided Sensing*: V2V communication enables “shadow perception,” where a vehicle with line-of-sight visibility can share its detection of occluded objects for other vehicles operating in visually impaired conditions [185], [186]. By tapping into this shared sensory data, multi-vehicle sensing strategies allow vehicles the potential to overcome their individual sensor limitations. For instance, LiDAR-equipped vehicles can broadcast 3D spatial maps to surrounding vehicles lacking such sensors, thereby extending the perception range and fidelity for non-equipped agents.

2) *V2I*: systems contribute an additional environmental context, such as traffic light state, pedestrian movement, or

weather conditions, which can be critical for nighttime safety [184]. Recent pilot deployments in urban V2X testbeds have demonstrated infrastructure-assisted perception improvements of up to 40% in object detection recall under night driving conditions [187]. V2X can allow broadcasting critical information for nighttime navigation, providing vehicles with updated environmental data and safety alerts [185]. This collaborative perception is particularly valuable in dark, congested urban settings, where improved situational awareness can reduce collision risks and increase the resilience of autonomous systems operating in low-visibility conditions.

The integration of smart infrastructure with onboard sensors supports faster response times, enhances collision avoidance, and improve overall safety for autonomous navigation, even in the dark. By combining data from infrastructure elements, V2I communication has the potential to enhance and expand the scope of perception.

3) *Complementary Fusion Strategies for Dark Environments*: Complementary data fusion plays a pivotal role in realising the full potential of cooperative DarkSensing. Furthermore, to fully capitalise on V2X communication for nighttime sensing, effective complementary data fusion techniques are essential. Integrating information from diverse sensors (*e.g.* LiDAR, cameras, infrared) across multiple vehicles and infrastructure sources can produce a more comprehensive and accurate environmental model. Fusion can occur at multiple levels: raw data, feature or decision level [185]. Approaches such as symmetric fusion (where all vehicles share and integrate multi-modal data equally) and asymmetric fusion (where one vehicle or server aggregates inputs) have been proposed in enhancing object detection, localisation, and situational awareness [111], [184], [185], [187]. These strategies can enhance performance in nighttime detection, lane estimation, and occlusion handling. New architectures such as CoBEVT [188] fuse multi-vehicle LiDAR and camera BEV features, demonstrating improved 3D object detection in low-visibility scenarios. These fusion approaches optimise perception by combining strengths across V2X data streams, enabling safer manoeuvring, accurate obstacle detection, and better-informed decision-making in challenging visual conditions.

4) *Challenges and Security Considerations*: Despite its promise, cooperative sensing introduces several technical challenges. Real-time bandwidth constraints, synchronisation between asynchronous sensors, and heterogeneity in sensor quality can affect the fusion reliability. Moreover, **cybersecurity** emerges as a key issue: adversarial spoofing or falsified data injection into cooperative streams could lead to incorrect perception or unsafe decisions [189]. Secure communication protocols and trust models are necessary to ensure robustness in dark environments. Additionally, cooperative datasets that explicitly include nighttime or low-illumination conditions remain scarce. Developing benchmark datasets for V2X in darkness is a critical need for advancing research.

VII. CASE STUDIES AND OPEN CHALLENGES

This section summarises the open issues and potential research directions in vehicular DarkSensing. A few relevant

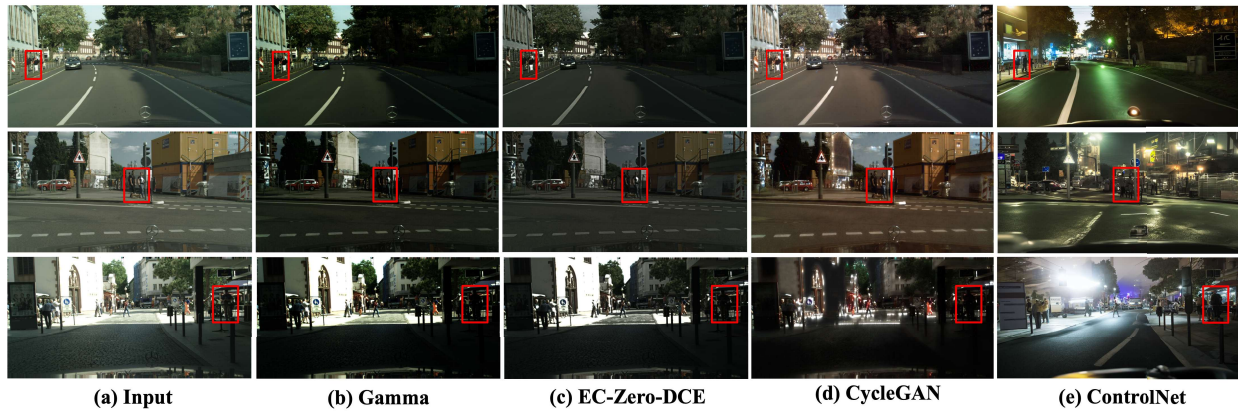


Fig. 6. Qualitative comparison of technologies to create dark simulated frames with lower illumination. From left to right: input normal light images, images augmented by: Gamma correction ($\gamma=2$), deep curve-based method EC-Zero-DC [190], the CycleGAN method [72], and the diffusion-based method [191].



Fig. 7. Qualitative comparison of current state-of-the-art image enhancement algorithms.

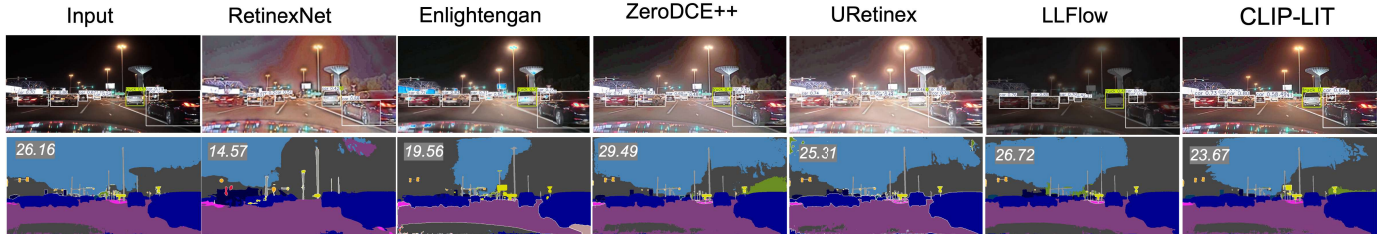


Fig. 8. Qualitative comparison of current SOTA low-light enhancement methods and downstream perception tasks [47], [58], [64], [192], [193]. The first row presents detection visualisation of the model predictions on the original nighttime input frame and enhanced output frames. The second row shows corresponding semantic segmentation results with their mean Intersection over Union (mIoU) scores on example frames. This example is from the Dubai scene in the NightCity dataset [14].

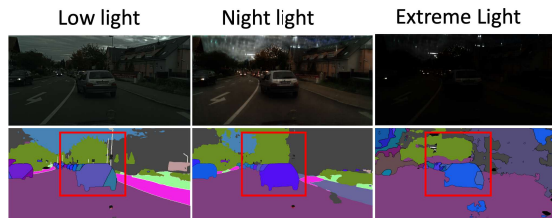


Fig. 9. Visual comparison of panoptic segmentation performance under varying dark conditions on the simulated D-Cityscapes+ dataset [8].

quantitative and qualitative experiments are used to support the discussion of the open challenges, as shown in Fig. 6–9, and Tab. VIII.

A. Dark Enhancement Related

1) *Real-World Generability Capability*: Although the current low-light image enhancement methods exhibit good

performance in brightening global illumination, problems still exist, such as overexposure of the output, with enhanced artefacts and noise. This issue results in limited generalisability when implemented in real-world driving scenarios, which have more complex and uneven artificial light sources. For instance, a method trained on the LOL dataset struggles to effectively enhance nighttime driving datasets [70], Fig. 7. In addition, as illustrated in Fig. 8, most enhancement methods (e.g. histogram-based, Retinex, GAN) fail to consistently improve perception performance in real-world night driving conditions. In some cases, enhancement introduces artefacts that reduce segmentation quality, highlighting the need for specific enhancement approaches for autonomous driving perception. Zero-DCE++ is the only model that shows an improvement in segmentation performance over the original input, highlighting the sensitivity of downstream tasks to the enhancement strategy. Furthermore, it is also a potential future research direction in investigating domain adaptation techniques to adaptively

TABLE VII

COMPARISON OF DARK-DRIVING DATASETS BASED ON SENSOR MODALITY, TIME-OF-DAY COVERAGE, WEATHER CONDITIONS, SUPPORTED PERCEPTION TASKS, LABEL GRANULARITY, AND NUMBER OF SEMANTIC CLASSES. D: DAYTIME, T: TWILIGHT, N: NIGHTTIME, /: NOT SPECIFIED

Dataset	Year	Data Format	Time of Day	Weather	Tasks(Classess)	Dark Scales
Raincouver [158]	2017	RGB	D, T, N	rain	semantic segmentation(3)	Train: 41; Test: 54
WildDash1 [159]	2018	RGB	D, T, N	mist, fog, rain, snow	semantic segmentation(28), instance segmentation(9)	/
Nighttime Driving [160]	2018	RGB	D, T, N		semantic segmentation(19)	Total: 9.5k; Test: 50
Dark Zurich [124]	2019	RGB	D, T, N		semantic segmentation(19)	Total: 2.6k; Val: 151
ExDark [5]	2019	RGB	D, N		object detection(12)	Train: 4.8k; Test: 2.6k
NightOwls [6]	2019	RGB	D, N	rain, snow	pedestrian detection(3)	Train: 128k; Val: 51k; Test: 103k
UNDD [161]	2019	RGB	D, N		semantic segmentation(19)	Total: 2.4k; Train: 35; Val: 10; Test: 30
BDD100k [162] (DarkBDD [163])	2020	RGB	D, T, N	fog, rain, snow, overcast, cloud	object detection(10), segmentation(19), tracking	Train: 27.9k; Val: 3.9k; Test: 8.0k
RA SSA [164]	2021	RGB	D, N	rain, cloud	semantic segmentation(8), domain adaptation	Total: 12.5k; Test: 226
NightCity [14]	2021	RGB	N		semantic segmentation(21)	Train: 3.0k; Test: 1.3k
ACDC [165]	2021	RGB	D, N	fog, rain, snow	object detection, panoptic, semantic(19) / instance(8) segmentation	Train: 400; Val: 106; Test: 500
WildDash2 [166]	2022	RGB	D, T, N	fog, rain, snow	panoptic, semantic(26), and instance(2) segmentation	/
Darkwick [167]	2024	RGB	D, N		low-light image enhancement	Total: 5.3k
KAIST [168]	2015	RGB, Thermal	D, N		pedestrian detection(3)	/
CVC-14 [169]	2016	RGB, Thermal	D, N		pedestrian detection	/
MF Dataset [170]	2017	RGB, Thermal	D, N		semantic segmentation(8)	Total: 749
Freiburg [134]	2020	RGB, Thermal	D, N		semantic segmentation(13)	Train: 8.6k; Test: 32
FLIR [171]	2021	RGB, Thermal	D, N	fog, rain, cloud	object detection(15)	/
LLVIP [101]	2021	RGB, Thermal	D, N		pedestrian detection	Total: 15.5k
M3FD [172]	2022	RGB, Thermal	D, N	fog, rain	object detection(6)	Total: 1.6k
OxfordRobotCar [173]	2017	RGB, LiDAR	D, N	rain, snow	localisation and mapping	/
A*3D [174]	2019	RGB, LiDAR	D, N	rain, cloud	object detection(7)	Total: 11.7k
Waymo [175]	2020	RGB, LiDAR	D, T	different weather conditions	3D detection(4), 2D object detection(3), tracking, segmentation	/
nuScenes [176]	2020	RGB, LiDAR, RADAR	D, N	rain, cloud	detection(23), tracking, segmentation(32)	Annotated: 4.6k
SHIFT [177]	2022	RGB, LiDAR	D, T, N	fog, rain, overcast, cloud	detection(6), segmentation(23), depth estimation	/
DDD17 [178], DDD20 [179]	2020	RGB, Event	D, N		event+frames fusion, steering prediction	/
MVSEC-NIGHTTL [180]	2021	RGB, Event	D, N		car detection	Train: 1.6k; Val: 400
LOD [181]	2021	RGB, Raw	N		object detection(8)	Train: 1.8k; Test: 400
ROD [182]	2023	Raw	D, N		object detection(6)	Train: 13k; Test: 1k
MUSES [9]	2024	RGB, LiDAR, RADAR, Event	D, N	fog, rain, snow	panoptic segmentation(19)	Total: 1k

TABLE VIII

COMPARISON OF PANOPTIC QUALITY METRICS IN D-CITYSCAPES+ DATASET UNDER LOW-LIGHT, NIGHT LIGHT AND EXTREME LIGHT

	Low-light	Night Light	Extreme light
P-Deeplab [194]	41.9	37.9	20.6
EfficientPS [195]	45.7	44.0	24.0
Oneformer [196]	63.4	60.2	52.5

enhance low illumination while intelligently removing noise. Region-specific enhancement techniques that can dynamically adjust enhancement levels based on local exposure conditions can be an option.

2) *Lack of Consideration of Temporal Information*: Few methods are designed for video enhancement, and most focus on spatial information without leveraging temporal information, which could be used as a clue during the enhancement process. Video enhancement algorithms that incorporate temporal information are a potential direction to improve the continuity and quality of enhancements across frames.

3) *Human Vision Vs Machine Vision*: Current image enhancement methods are mainly designed to improve human visual quality, which does not necessarily translate to improved machine vision quality. For example, a higher PSNR in enhanced images does not necessarily indicate a better detec-

tion or segmentation accuracy [8]. Shifting the focus towards enhancement techniques that optimise for machine vision performance, possibly by developing metrics that better correlate with machine vision tasks, is a crucial direction for future research.

B. Dark Perception Related

1) *Unrealistic Methods to Generate Synthetic Data*: There is a lack of realistic dark-driving data synthesis methods, which can result in the wrong supervision of image-to-image translation in night perception methods [117]. A qualitative case study is presented in Fig. 6, with both conventional and DL-based synthetic methods. The Gamma translation is the most stable one with the ability to darken with different levels. The curve-based EC-Zero-DCE method shows a similar effect. However, they cannot simulate the presence of diverse nighttime light sources. While GAN-based methods show a more diverse light distribution, they exhibit several limitations: (a) improper darkening: not adequately reducing illumination levels and abnormal light generation in illogical locations. (b) dark artefacts: unnatural illumination effects throughout the image, loss of small or far objects. The diffusion-based model has the ability to generate more diverse light colours; however, there are problems in controlling the structure of the inputs. Improving synthetic data generation techniques

to produce more realistic nighttime images or other sensor data, possibly incorporating physics-based models to control dark levels while maintaining the unique character of various classes, is an important area for future exploration.

2) *Compound Noise Factors and Tasks*: While most current models address low illumination in isolation, real-world nighttime scenarios often include other complicating factors, such as rain, fog, glare, and wet road surfaces. These elements can cause reflections, motion blur, and occlusions, degrading sensor reliability and reducing perception accuracy. For example, fog can scatter light and diminish LiDAR return power, while wet roads can reflect headlights, confusing vision-based systems. The MUSES dataset was recently proposed to discuss the panoptic segmentation performance in both day and night with different weather conditions, where the paper shows the degradation in various weather conditions at night (e.g. 20% drop in panoptic quality (PQ) from day to night, with 42.2 PQ in snow and 46.5 PQ in fog) [9]. Developing multi-noise robust architectures or weather-aware perception models is crucial for safe deployment. Furthermore, dark-scene datasets that combine nighttime with diverse weather conditions remain limited, and more benchmarks are needed to facilitate research in this space. In addition, most of the current dark perception tasks are for detection or segmentation; emerging techniques in relatively new perception tasks, such as tracking or planning for DarkSensing, can be developed in the future.

3) *Extreme Dark Challenges*: In extremely dark environments, the photons captured by camera sensors are limited. Due to the limited information, current enhancement methods could fail in restoring the structural details of visible images [13]. As illustrated in the case study from Fig. 9 and Tab. VIII, a significant degradation is shown in both visual and quantitative segmentation accuracy under extremely low-light conditions, with more than a 50% PQ drop compared to low-light. Interestingly, while all backbones suffer, transformer-based architectures retain relatively higher robustness. This highlights the growing potential of transformer-based perception models for DarkSensing tasks, especially for extreme light challenges.

C. Dark Sensor Related

1) *Lack of Proper Datasets*: For dark enhancement, although most current paired normal-light and dark datasets are synthetic or static indoor datasets, they have a significant domain gap compared to real driving paired daytime and nighttime datasets. For dark perception, there is a limited annotated nighttime driving dataset with multi-modality, leading to the popular usage of the unsupervised-based method, which may fail if the domain gap is substantial. Creating and annotating large-scale paired driving datasets or multi-modal datasets that include a wide range of lighting conditions is essential to facilitate better training and evaluation of DarkSensing models [167]. The exploration of semi-supervised techniques to bridge domain gaps is also a necessary step forward.

2) *Balancing Cost and Sensor Modality*: As the information captured by the visible images is the most commonly used perception sensor in AAD, additional guidance from other sensor data modalities (e.g. thermal, point cloud, event

data) can support the enhancement or the perception process. Most of the off-the-shelf cross-modal feature fusion methods are designed for specific sensors without considering the inherent performance of the sensors' in the dark. Conducting cost-benefit analyses and developing lightweight and robust sensor fusion approaches to balance performance and practical deployment constraints will be essential for future research.

3) *Deployment Challenges in Real-World Cooperative DarkSensing*: While cooperative perception offers great potential in low-illumination scenarios, several practical challenges limit its large-scale deployment in real-world autonomous systems.

a) **Communication latency and bandwidth constraints** are particularly acute in dark sensing scenarios due to the reliance on either longer post-processing time (e.g. dark enhancement algorithm) or higher-density and higher-resolution sensor data (e.g. LiDAR point clouds, infrared imagery) necessary for effective nighttime perception. Real-time response requirements might be increased. These pipelines demand low-latency and high-bandwidth V2X communication to transmit detailed sensor data without delays that compromise immediate decision-making [197]. In dark conditions, limited bandwidth or intermittent connectivity exacerbates the risk of incomplete or delayed sensory information, increasing vulnerability to hazardous manoeuvres due to compromised situational awareness [15]. Future advancements such as cooperative perception leveraging optimised V2X communication protocols and decentralised edge-based processing could have great potential to mitigate these issues, enhancing vehicle collaboration to maintain accurate object tracking and perception integrity under darkness-specific visibility challenges.

b) **Security and trustworthiness** of data exchanged in dark scenarios are particularly sensitive, given the potentially increased reliance on cooperative perception to overcome low visibility. Malicious interventions, such as the spoofing of sensory data or the injection of false objects, can severely disrupt vehicle planning decisions under low-light conditions, where autonomous systems already face higher uncertainties [189]. Robust authentication mechanisms, rigorous data validation protocols, and enhanced cryptographic methods specifically designed to safeguard nighttime cooperative sensing frameworks are essential to prevent adversarial manipulations that exploit the inherent visibility limitations of nighttime environments.

c) **Infrastructure limitations** could significantly constrain cooperative sensing during nighttime, when vehicular perception depends heavily on infrastructure-supported sensing aids, such as smart street lighting, edge-computing nodes with enhanced nighttime processing capabilities, and advanced roadside units (RSUs) equipped with night-specific sensors (infrared, thermal). The absence or insufficient deployment of such specialised infrastructure exacerbates visibility constraints faced by vehicles, particularly in rural, suburban, or underdeveloped regions [184]. Additionally, the requirement for precise synchronisation, standardised data formats, and calibration of diverse sensors is intensified in

low-visibility conditions, complicating cooperative sensing integration among heterogeneous fleets in dark conditions.

d) Dataset and benchmarking gaps further challenge development. There is a lack of open, annotated datasets that simulate V2X-based cooperative perception in dark conditions. Without such datasets, algorithm development, evaluation, and reproducibility become difficult. *Addressing these deployment challenges will require interdisciplinary collaboration across networking, security, systems engineering, and autonomous driving to ensure that cooperative DarkSensing moves from theoretical promise to practical reality.*

4) *Human Factors and User Acceptance:* Beyond technical challenges, understanding drivers' subjective experiences in low-light scenarios is critical for the development of more human-aware dark sensing systems. A study by Ozturk and Merat explored how self-perceived driving abilities influence people's experiences during night driving [198]. Their findings revealed that young drivers who rated themselves lower in perceptual-motor skills but higher in safety-related behaviours were more likely to report difficulties when driving at night. This finding suggests a complex interplay between self-confidence in vehicle handling and cautious driving tendencies, which may heighten the perceived challenges of low-light conditions [198]. Another behavioural study emphasised how road sign and marking visibility can subtly influence eye movement and cognitive load, with better visibility leading to increased driving speeds without significant changes in lateral control or deceleration [199].

Other research emphasises the role of infrastructure in shaping driver behaviour at night. For instance, well-placed intersection lighting was found to reduce crash risks in a simulation-based study, while reduced headlight intensity in well-lit urban areas may reduce glare without compromising detection [200], [201]. Tunnel design that increases background luminance has also been shown to mitigate fatigue and improve attentiveness [202]. He et al. found that anger led drivers to accept shorter gaps and take greater risks during left turns at night [203]. Similarly, cognitive distraction in the form of multitasking (e.g. n-back tests) was found to reduce peripheral detection performance during nighttime driving, especially in older drivers, which could pose safety risks for vulnerable road users like pedestrians [204]. Despite these findings, there is a lack of broader research into public perception and acceptance of DarkSensing technologies. More studies are needed to explore how drivers evaluate and trust these systems, particularly in real-world contexts. Understanding user expectations will be critical for ensuring that technological advancements align with diverse road users' needs and comfort levels.

VIII. CONCLUSION

In this paper, we have conducted a comprehensive review of DarkSensing, focusing on the critical aspects of vehicular sensing in the dark (*i.e.* low-light, nighttime and extreme dark) from both hardware and software perspectives. Within intelligent transportation systems, DarkSensing has the potential to contribute to overall safety, robustness, and reliability.

The impact of DarkSensing extends beyond individual vehicle perception, influencing cooperative driving, infrastructure coordination, and broader connected mobility systems. The successful deployment of DarkSensing depends on the integration of advanced low-light sensor technologies, illumination-robust algorithms, and cooperative perception frameworks that align with the multi-stage pipeline of autonomous systems. Despite rapid progress, several research gaps remain. For example, the lack of standardised dark benchmarks, limited multimodal sensor fusion, and the absence of research on human factors, robust and secure V2X frameworks tailored for the dark conditions are discussed. Addressing these challenges will require interdisciplinary efforts spanning sensing, AI, and communication protocols. This convergence of sensing, learning, and cooperation in dark scenarios lays the foundation for safer, more resilient, and context-aware autonomous mobility. Looking forward, DarkSensing holds the potential to not only improve nighttime driving safety but also influence policy, urban infrastructure planning, and the design of future intelligent mobility ecosystems.

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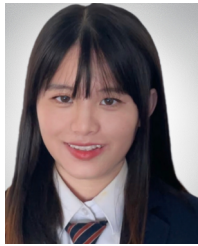
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Yiting Wang (Graduate Student Member, IEEE) received the master's degree (Hons.) in smart connected and autonomous vehicles from the University of Warwick, U.K., where she is currently pursuing the Ph.D. degree with WMG, IV Sensors Group. Her Ph.D. study is funded by CDT. She was a Research Intern at the Laboratory of Artificial Intelligence Thrust, HKUST, in 2022. Her research interests include intelligent vehicles, perception robustness, data degradation, and simulation. She is an AAAI PC Member. She serves as a reviewer for journals and conferences, such as IEEE TRANSACTIONS ON INFORMATION THEORY, IEEE TRANSACTIONS ON INTELLIGENT VEHICLES, and IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS.



awarded the State Scholarship Fund by China Scholarship Council.

Boda Li received the M.Sc. degree in sustainable automotive engineering and the M.Sc. degree in smart, connected, and autonomous vehicles from Warwick Manufacturing Group (WMG), University of Warwick, Coventry, U.K., where he is currently pursuing the Ph.D. degree with the Intelligent Vehicle Sensors Group. His research interests include automotive camera corner cases and noise factors, machine learning, perception sensors, and assisted and automated driving. He is an Associate Fellow of the Higher Education Academy and has been



Zixiang Wei (Graduate Student Member, IEEE) is currently pursuing the Ph.D. degree with the IV Sensor Group, University of Warwick. He was a Research Assistant at The Hong Kong University of Science and Technology, the VLIS Laboratory of Artificial Intelligence Thrust, and the Department of Electrical Engineering, Tongji University. His research interests include low-level computer vision tasks. His current research interests include data degradation and its impact on learning based models. He has served as a reviewer for IEEE

TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, *PeerJ Computer Science*, and *IET Image Processing*.



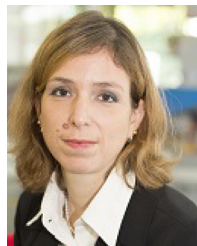
Anima Rahman received the B.Eng. degree in electronic and electrical engineering from the University of Birmingham and the master's degree in automotive technology, with a focus on automotive sensors, sensor fusion, machine intelligence, and human-technology interaction. She is currently pursuing the Ph.D. degree in future mobility technologies with WMG, University of Warwick, with a focus on intelligent vehicle sensors and driver monitoring systems. Her research interests include machine learning to develop non-contact solutions for driver state detection.



Daniel Gummadi received the Master of Engineering (M.Eng.) degree in systems engineering from the University of Warwick, U.K., in 2022. From 2022 to 2024, he was a Research Engineer with the Intelligent Vehicles Group, WMG, University of Warwick. During this time, he contributed to the EU ROADVIEW Project, focusing on the development of physics-based modeling of noise factors in automotive sensors, as well as the EU Hi-Drive Project and the Innovate U.K. Sim4CAMsens Project. His research at WMG centred on sensor data capture, noise modelling, data validation, and the impact of sensor noise on perception systems. He is currently an Autonomous Vehicles Engineer with Oxa. His research interests include perception sensors, intelligent vehicles, and computer vision.



Haonan Zhao received the bachelor's degree from the Department of Vehicles, Tongji University, China, and the master's degree (Hons.) from the University of Warwick, U.K., where he is currently pursuing the Ph.D. degree from the Visualisation Group, WMG. His research interests include data synthesis, perception, and generative AI.



Valentina Donzella (Senior Member, IEEE) received the B.Sc. degree in electronics engineering from the University of Pisa, Pisa, Italy, in 2003, and the M.Sc. degree in electronics engineering and the Ph.D. degree in Innovative technologies for information, communication and perception engineering from the Sant'Anna School of Advanced Studies, Pisa, in 2005 and 2010, respectively.

In 2009, she was a Visiting Graduate Student at the Engineering Physics Department, McMaster University, Hamilton, ON, Canada. She is currently a Full Professor, the Head of the SPRING Group, School of Engineering and Materials Science, Queen Mary University of London, U.K. She is also leading the work package on perception sensor noise models as a part of the four-year EU ROADVIEW Project. Previously, she was an MITACS and SiEPIC Post-Doctoral Fellow at the Silicon Photonics Group, The University of British Columbia, Vancouver, BC, Canada. She is the first author, co-author, and last author of several journal articles on top-tier optics and sensors journals. Her research interests include LiDAR, intelligent vehicles, integrated optical sensors, sensor fusion, and silicon photonics. She is a Full College Member of EPSRC and a Senior Fellow of the Higher Education Academy. She was awarded the Royal Academy of Engineering Industrial Fellowship on camera sensors from 2020 to 2022. She is the current Co-Chair of the IEEE ITSC UK Chapter and an Associate Editor of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS.